

What are the various shell types?

The following are the various types of shells, explained in brief.

sh: The Bourne shell (sh) or command line interpreter is for computer operating systems. This was the default shell for Version, in earlier variants of UNIX 7.

In Ubuntu, this is also called 'dash'. This is a POSIX-compliant implementation of `/bin/sh` that aims to be as small as possible. 'dash' is a direct descendant of the NetBSD version of 'ash' (the Almquist SHell), ported to Linux in early 1997. It was renamed *dash* in 2002. The following command will take the reader to the detailed man page for sh in Ubuntu:

```
ubuntu@ubuntu:~$ man sh
```

 **Note:** 'sh_shell_screen_shot' will give a detailed view of the output of this command.

Bash: Bash is a UNIX shell and command language written by Brian Fox for the GNU Project as a free software replacement for the Bourne shell.

```
ubuntu@ubuntu:~$ man bash
```

csh: The C shell is a UNIX shell created by Bill Joy, while he was a graduate student at the University of California, Berkeley in the late 1970s. This shell was widely used by the C-programming community earlier.

Ksh: KornShell is a UNIX shell and was developed by David Korn at Bell Labs in the early 1980s. It was based on the Bourne shell source code. This shell was the default in many proprietary UNIX flavours like AIX, HP-UX, etc.

 **Note:** In some of the Ubuntu flavours, this will not be available, by default. We need to install it using the `sudo apt get install ksh` command.

Pdksh: The Public Domain Korn Shell, also known as `pdksh`, is open source and is used predominantly in UNIX/Linux like systems.

Tcsh: This is an enhanced but completely compatible version of the Berkeley UNIX C shell, `csh(1)`. It is a command language interpreter usable both as an interactive login shell and a shell script command processor.

User details

The following are the well-known and popular user detail based commands.

1) **whoami:** This command is used to print an effective user ID.

```
ubuntu@ubuntu:~$ whoami
ubuntu
```

2) **uname:** This command is used to print system information. There are many options, but '-a' is most

popular and will display all the information.

```
ubuntu@ubuntu:~$ uname -a
Linux ubuntu 4.10.0-19-generic #21-Ubuntu SMP Thu Apr 6
17:04:57 UTC 2017 x86_64 x86_64 x86_64 GNU/Linux
```

3) **who:** This shows who has logged in with terminal information.

```
shashi-ubuntu@ubuntu:~$ who
ubuntu tty7          2018-04-09 09:09 (:0)
```

Directories

There are seven to eight directories present across all the UNIX/Linux flavoured OSs.

Some of the following are popular and it is best if users get familiar with them.

- bin:** All the binary executables are present in this directory.
- lib:** All the library files are present here.
- dev:** All the device files (character, block, socket, etc types) are here.
- etc:** This is for all systems admin related commands.
- usr** and **/usr/bin:** This is the home for users and additional executables.

Creating and viewing files, and copying/removing commands

There is a long list of commands used for creating, viewing, copying and removing files and file operations. Here, I have listed the most popular ones.

a) **touch:** This command is used to create new files and to change timestamps.

```
ubuntu@ubuntu:~$ touch 123
ubuntu@ubuntu:~$ ls -l
total 44
-rw-r--r-- 1 ubuntu ubuntu  0 Apr  9 09:36 123
drwxr-xr-x 2 ubuntu ubuntu 4096 Mar 24 09:52 Desktop
drwxr-xr-x 2 ubuntu ubuntu 4096 Mar 24 09:52 Documents
drwxr-xr-x 2 ubuntu ubuntu 4096 Mar 24 09:52 Downloads
-rw-r--r-- 1 ubuntu ubuntu 8980 Mar 24 09:42 examples.desktop
drwxr-xr-x 2 ubuntu ubuntu 4096 Mar 24 09:52 Music
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr  6 04:44 Pictures
drwxr-xr-x 2 ubuntu ubuntu 4096 Mar 24 09:52 Public
drwxr-xr-x 2 ubuntu ubuntu 4096 Mar 24 09:52 Templates
drwxr-xr-x 2 ubuntu ubuntu 4096 Mar 24 09:52 Videos
```

- cat:** This concatenates files and prints them on the console. The `cat` command comes up with a lot of options, including re-directions (using operators like `<`, `>`, `<<`, `>>`, etc).
- cp:** This command is used to copy files and directories.

```
ubuntu@ubuntu:~$ cp 123 321
```